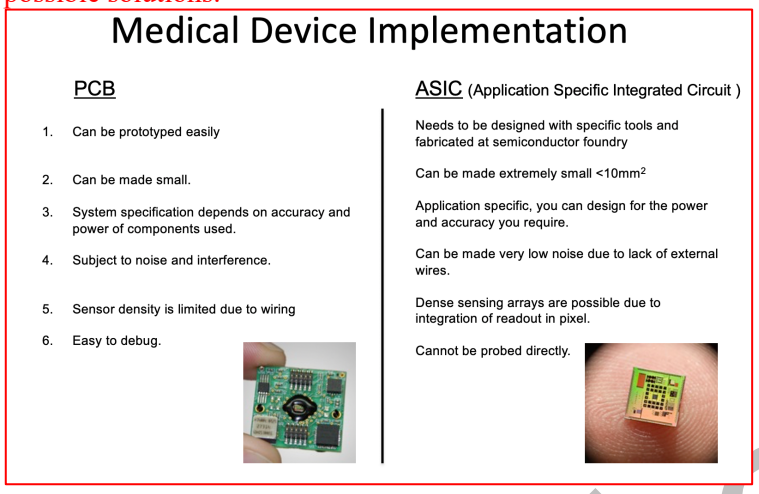
1. This question relates to the design of a multichannel instrument for *measuring bio-potentials from muscles* located on the back. The instrument will use a 5-channel electrode system shown in Figure 1.1 and the specifications of commonly observed bio-potentials is shown in Table 1.1.

   a) The system can be implemented on a PCB with external components or an Application Specific Integrated Circuit (ASIC) in CMOS. Explain the advantages and disadvantages of each and state which you would use and why?

Student needs to give his own perspective based on the advantages and disadvantages of each. Both are possible solutions.



[5]

   b) Design a complete instrumentation system for recoding bio-potentials from the back according to the following specifications:

Instrumentation Amplifier Gain: 20
Low-Pass Filter Roll off: 60dB/decade
High-Pass Filter Roll off: 40dB/decade
Total number of op-amps available for each filter: 1
Supply Range: -1.8V to 1.8V

State all assumptions and values of resistors and capacitors used. You may assume that the instrumentation system utilizes the whole dynamic range available and 1nF, 10nF and 100nF capacitors are available for the filters.

Using table below for EMG:
High pass cut off: f_{HP}=20Hz
$R_{HP}=1/2\pi Cf_{HP}$ = 1/2π(10nF)(20Hz) =0.795 MΩ [1 point]
Low pass cut off: f_{LP}=2KHz
$R_{LP}=1/2\pi Cf_{LP}$ = 1/2π(10nF)(2KHz)=7.95KΩ [1 point]
Note, needs to be scaled by a factor of 10 for each non buffered stage.

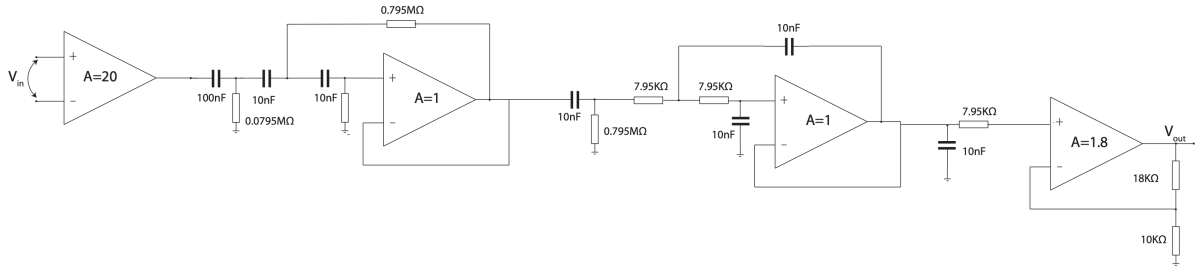
2 opamps for filters.
Low pass 60dB/decade: Sallen Key can be used but RC l in input and output.
High pass 40dB/decade: RC + Sallen Key [1 point]

Instrumentation amplifier gain:
$V_{OUT} = -(V_1 - V_2)(1 + 2R_2/R_1)(R_4/R_3)$
For gain of 20 we select R4=R3=R2= 95 KΩ and R1= 10 KΩ [1 point]

Supply Range: -1V to 1V
VDD=1.8V, GND=0V, VSS=-1.8
Max EMG = 100mV (from table)
V_{inptp} x A1 x A2=3.6V_{ptp}

Solutions

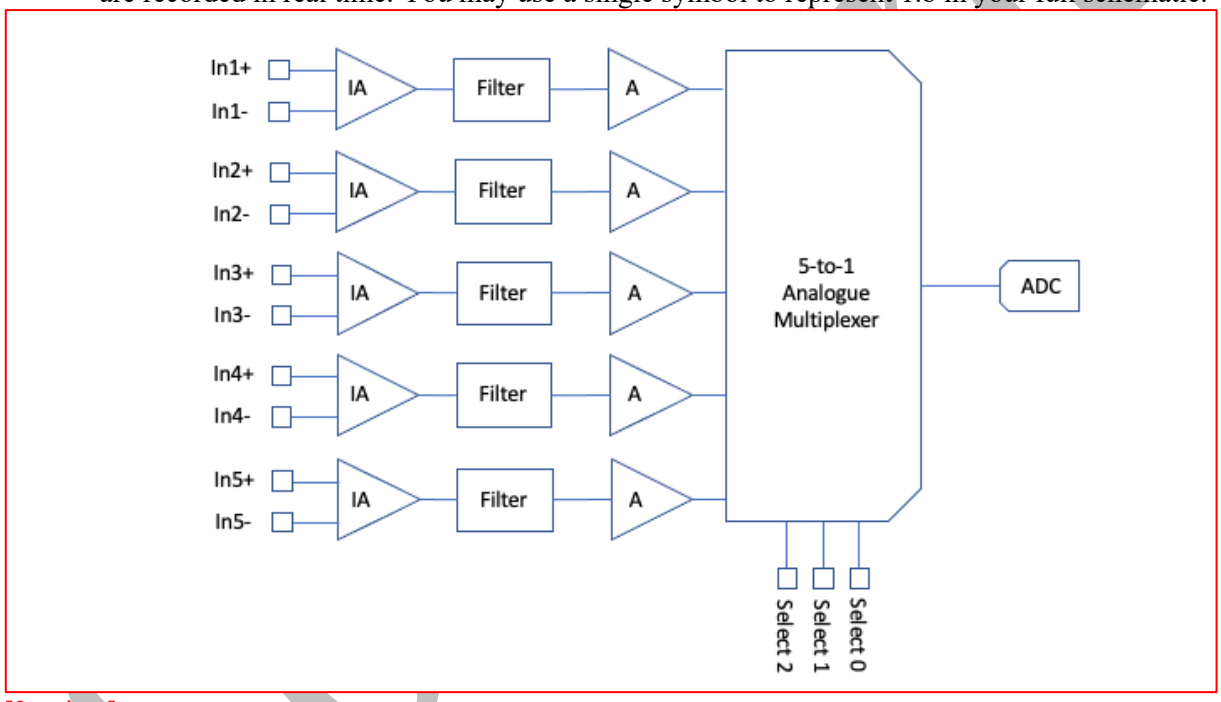
A1=20
100mVx20xA2=3.6V
A2= 1.8 (gain of filter and final stage) [1 point]
Realised with a non-inverting amplifier with R1=18KΩ and R2=10KΩ [1 point]



[5 points]

[10]

c) Using your design from 1.b, sketch the schematic of the complete instrument assuming your system has only one ADC output and state any timing considerations to make sure all 5 channels are recorded in real time. You may use a single symbol to represent 1.b in your full schematic.



[3 points]

Timing considerations: 5 Channels means 3 select bits.
Each channel requires 4Khz sampling assuming Nyquist.
Clock frequency must therefore be a minimum of 5 times that, 20Khz.
Therefore, ADC will be time interleaved working at a minimum of 20KSamples/sec
[2 points]

[5]

Table 1.1

Signal	**Frequency range (Hz)**	**Amplitude range (mV)**
ECG	0.1 – 300	0.05 – 3
EEG	0.5 – 40	0.001 – 1
EMG	20 – 2000	0.001 – 100
Neural Spikes	300-5000	0.001– 0.5
Local Field Potentials	10-200	0.001 – 5

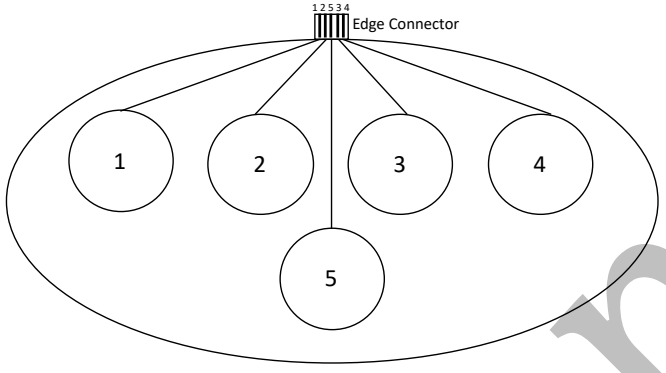


Figure 1.1

2. This question relates to electrical stimulation applied to sensory prostheses:

   a) Describe the relative merits of different electrode configurations (e.g. monopolar, bipolar, tripolar, multipolar). Why do most clinical devices utilise monopolar configurations? [5]

Multipolar strategies can be employed to "focus" the electric field, and thus improve selectivity. For example, in a bipolar configuration a pair of electrodes is used for each target site/confining field to tissue between that electrode pair. (2 marks)

Monopolar configuration on the other hand has a single electrode per site and a common return electrode. The activated tissue is therefore less well defined compared to bipolar or multipolar configurations. (1 mark)

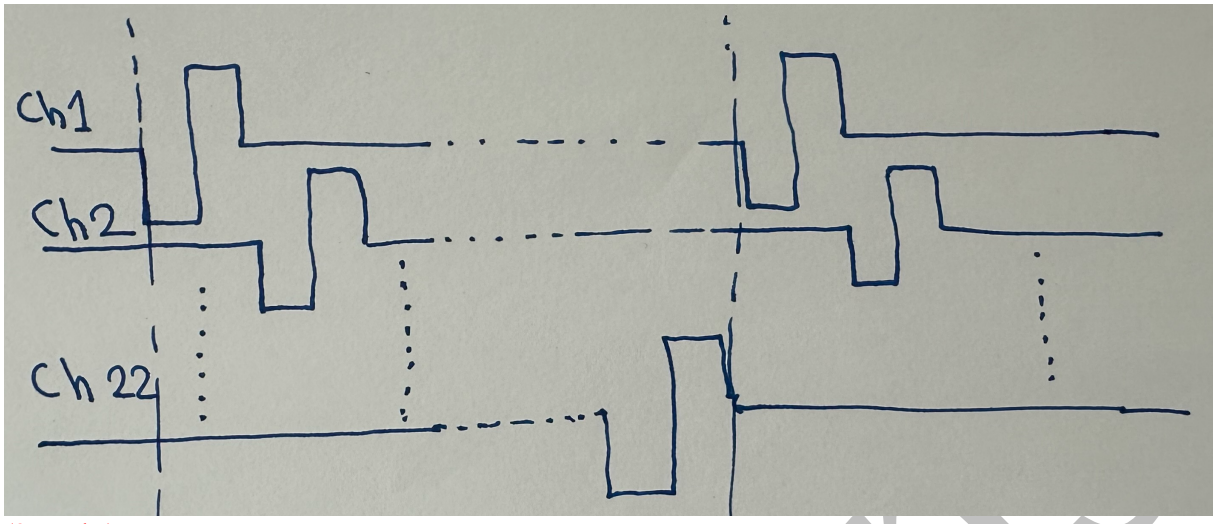
Most clinical devices, for example cochlear implants use simple monopolar configurations as this provides best "efficiency" for activating target volume of tissue with minimal stimulation current (therefore power). (2 marks)

   b) Explain why it is challenging to apply electrical stimulation to applications requiring high resolution, for example, a retinal prosthesis for restoring vision. [5]

The main challenge stems from fact that it is difficult to control the spread of electrical stimulation in tissue. (1 mark) High resolution means a large number of sites, which means the electrode size needs to be small – this bring challenges with electrode materials (each material has a maximum safe charge density it can support). (1 mark) Also given the challenge with current spreading it is challenging to do concurrent (parallel) stimulation across multiple sites, without having crosstalk between the different electrode pairs (1 mark). Therefore, any stimulation needs to be time division multiplexed, meaning any high-resolution applications such as retinal prosthetics have limitations to temporal update (e.g. frames per second). (2 marks)

Solutions

c) Sketch a biphasic waveform suitable for delivering 30nC/phase stimulus for a 22-channel cochlear implant. Annotate magnitudes and timings on your sketch. State any assumptions made. [5]



(2 marks)

Assuming the refresh rate is say 1kHz, within 1ms there needs to be 22 stimulation cycles (for a CIS strategy), therefore 45us per cycle = 45us/2=22.5us per stimulation pulse. (2 marks)

For stimulus of 30nC, Istimulus = 30nC/22uS=1.4mA. (1 mark)

d) (i) Estimate the overall power consumption of the cochlear implant assuming an average stimulus as above. [4]

Assuming average stimulus of 1.4mA (as per part (c)), and electrode resistance of 10kohms, 14V voltage drop across tissue is required, assuming another 3V for circuits, total power supply = 17V. (1 mark)

Therefore stimulation power consumption = 17V x 1.4mA = 23.8mW. Assuming overall system efficiency of 20%, average power consumption for complete device would be 119mW. This would be when maximum stimulus is delivered continuously. Assuming the average is 10% the maximum stimulus, average power consumption = 11.9mW. (3 marks)

(ii) How long would a PowerOne Implant battery last (Zinc air, 1.4V, 640mAh)? [1]

Thus for 640mAh x 1.4V => This would last about 640*1.4/11.9 = 75 hours. Assuming 15hrs/day usage (about 5 days). (1 mark)

Solutions

3. Figure 3.1 shows a smart contact-lens system for non-invasive measurement of tear glucose. The glucose concentration in the tear fluid is within 0.1-0.6mM for a healthy person and may go up to 1mM in patients with diabetes. The Glucose sensor comprises a working electrode (WE) with an area of 1mm^2, a reference electrode (RE) (area:0.3mm^2) and a counter electrode (CE) (area:2mm^2). The sensor readout IC receives AC power from the antenna and sends the measured data back to the transmitter. The instrumentation operates on a single supply with a VDD of 1.8V.

The working electrode is coated with glucose oxidase to give a peak current to glucose at V_{Cell}=0.7 V according to the following reaction:

$$\text{Glucose}+O_2 \xrightarrow{Glucose\ Oxidase} \text{gluconic acid}+ H_2O_2$$
$$H_2O_2+O_2 \xrightarrow{V=0.7V} 2H^+ + 2e^-$$

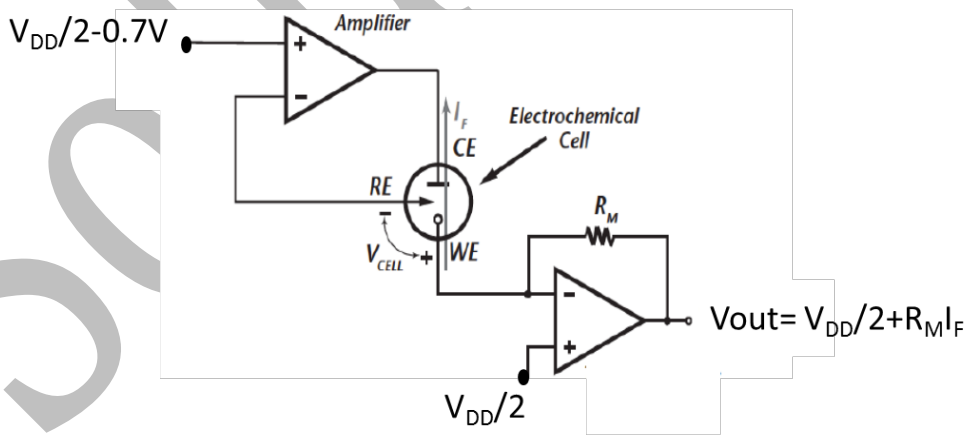
a) Calculate the peak redox current for the reaction above using the Randels Equation below. You may assume the number of electrons *n* is solely given by the oxidation of H_2O_2 , the scan rate is 1V/sec and the diffusion coefficient D=4.38X10^{-8} m^2/s:

$$i_p = 2.69x10^5 \cdot n^{3/2} \cdot A \cdot D^{1/2} \cdot C \cdot v^{1/2}$$

n=2
A=1mm2
D=4.38X10^{-8} m2/s
C=1mM
V=1V/sec
I= 2.69*10^5*(2)$^{1.5}$*(10^{-6})*(4.38X10^{-8})$^{0.5}$*(10^{-3})*(1)$^{0.5}$= 159 nA

[2]

b) Design a potentiostat and trans-impedance amplifier (TIA) with opamps and resistors to apply the cell voltage, measure the sensor current and maximize the output swing. Annotate the schematic.



[2]

V_{cell}= 0.7 V
VDD= 1.8V, $I_{F,max}$= 159nA
R_M*$I_{F,max}$=< 0.9V
Then R_M =< 5.66 MOhm

[2]
[Total 4 points]

[4]

c) Describe two alternative circuits to measure the sensor current, and state one advantage and one disadvantage for each with respect to TIA in part 3.b.

Solutions

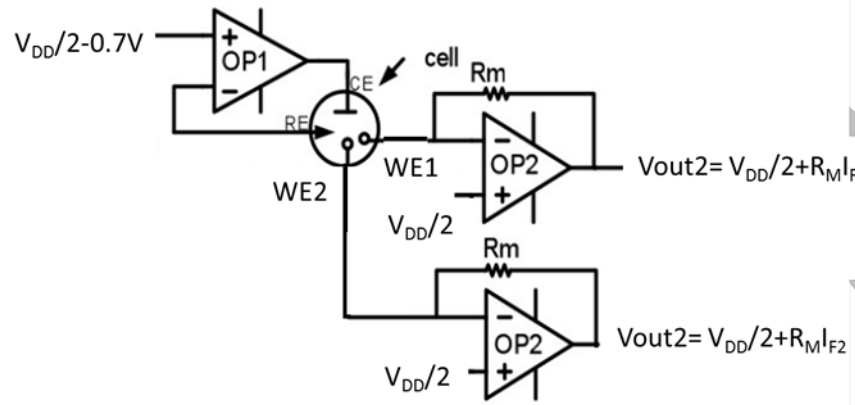
Switch capacitor based TIA; adv: can be implemented on-chip as doesn't need big resistor [1]. Disadvantage: Reliant on timing so it is Nyquist limited.
Grounded WE with IA measurement on CE: adv: We electrode is grounded so less noisy. [1]. Disadvantage: Needs more opamps to bootstrap.

[Total 4 points]

[4]

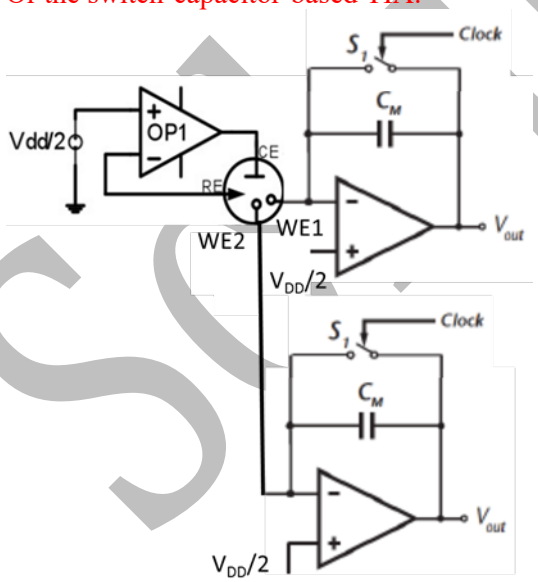
d) Common mode sensor interference and current drift limits the performance of bio-sensing systems. Propose one method that can help to compensate these effects and redesign the circuit in part 3.b and one of the circuits in part 3.c to implement this method.

Differential measurement can be implemented to reduce the effect of drift and interferences. This can be answered with any of the resistor-mode TIAs, SC-TIA or current-mode potentiostat. The two resistors based TIA circuit :



[2]

Or the switch-capacitor-based TIA:



[2]
[Total 4 points]

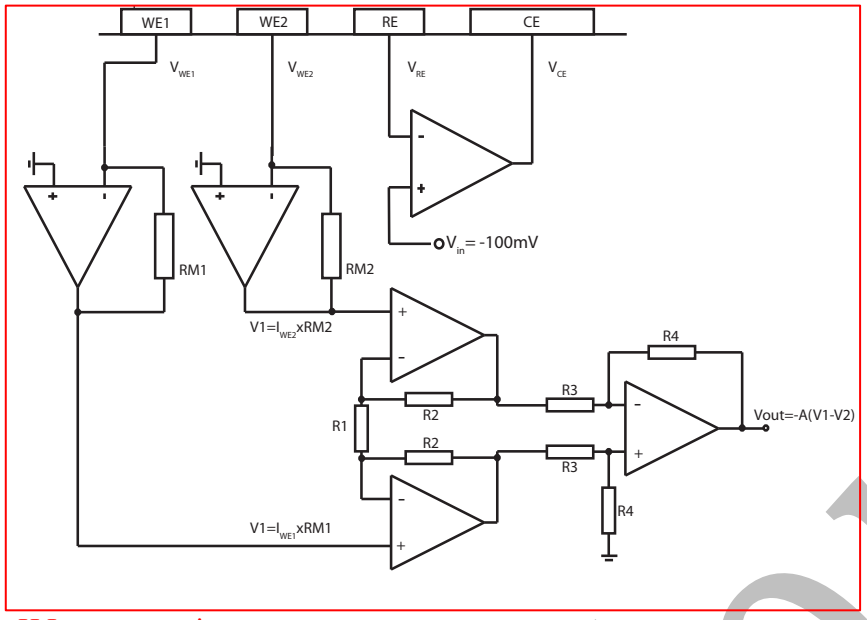
[4]

e) The tear glucose measurement has a non-linear dependance on pH. This however can be compensated with the addition of a sensor to measure pH. Design a complete front end for the smart-contact lens which includes a glucose and pH measurement but also compensation for common mode sensor interference and drift. You only have one ADC to sample both channels.

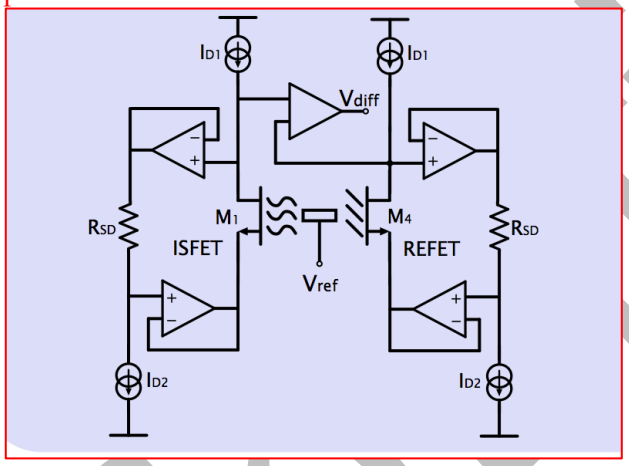
Scanned

Student need to identify the use of the ISFET and include the following instrumentation with for the differential measurement in both arms:

Glucose instrumentation:



pH Instrumentation:



Then output is connected to a 2-bit Analogue MUX and ADC.

[6]

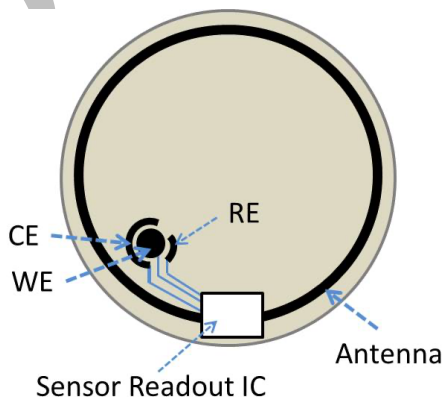


Figure 3.1

Solutions

4. This question relates to implantable brain computer interface (iBCI) devices:

a) What are the advantages/disadvantages of an intravascular BCI compared to other iBCIs? [5]

Advantage of intravascular BCI is it does not require any brain surgery, or opening in the skull to insert the implant, it is inserted through a blood vessel via the jugular vein [1].

Disadvantage is the electrodes are placed in a blood vessel, so not in direct contact with the neural cells (neurons) whose activity is being observed, but at some distance. As such, one my expect to only observe local field potential activity and not the more informative spiking activity. (2 marks) Also the precise target (location of the electrodes) is limited by where the blood vessel is – this might not be at the desired location, to the relevance of the activity/data may not be perfect. (1 mark) Finally, as the electrodes are some distance from the tissue, the maximum resolution (number of electrodes) would also be limited, again limiting the amount of useful information that could be extracted with such a device (1 marks).

b) Estimate the amount of raw data (kbps) generated by this iBCI (illustrated in Figure 4.1). [5]

If considering a 16-channel device that is observing only field potential data (1 mark). If LFP bandwidth is 100Hz (1 mark), assuming sampling frequency >2x Nyquist, say 1kS/s (1 mark), with each sample having 16-bit resolution (1 mark). Total data rate thus = 16 x 1000 x 16 = 256kbits/s or 32kbytes/s (relatively low datarate) (1 mark).

c) Explain how information is extracted from the low-level data received by the computer. [5]

Typical strategies for decoding motor activity are based on decoding an end-point (position) a velocity, or goal (1 mark) The raw data is typically filtered (to remove any high frequency noise, as we wouldn't expect to be using spike activity), then passed through bandpass filter bank and power detected in different frequency bands (2 marks). These signals (i.e. 16 channel x 8 bands) are used as the inputs to a decoder (1 mark). The actual decoder can be a population vector machine, a kalman filter, optimal linear estimator, or be based on DL methods (e.g. neural networks) (1 mark).

d) In what ways could such an iBCI be improved? [5]

Key issues to be improved would be everything mentioned above as a disadvantage. Invasiveness is better than intracortical iBCI but could be completely wireless (remove need for wires to remain in blood vessel) (1 mark).

On issue with data quality/decoding quality- being far from neural tissue/etc. More channels (1 mark), extended reach (more regions) (1 mark), new decoding strategy (try to exploit control signals in brain regions directly adjacent to electrodes) (1 mark), combine neural signals with other sensors (e.g. external eye tracker) (1 mark)

Solutions

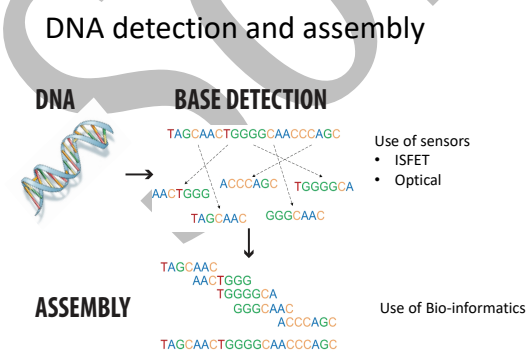


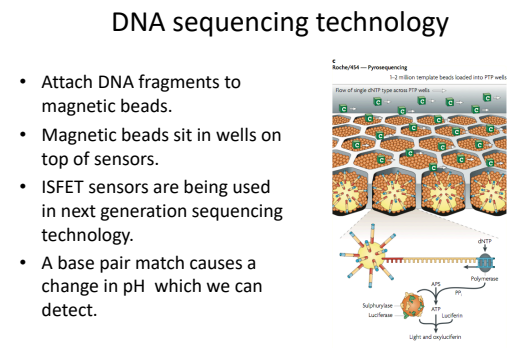
Figure 4.1. an intravascular iBCI that is currently in human trials. This utilises a 16-electrode array inserted in the superior sagittal sinus blood vessel.

5. This question relates to the design of an instrument for conducting DNA sequencing.

   a) Explain the process of DNA sequencing describing the end-to-end procedure from an unknown strand of human DNA and the sensing mechanism used.

Take the human genome under investigation.
Chop the 3.2 Billion base pairs into strands of 100-1000 bases with overlaps. These are called fragments.
Immobilise the primers on magnetic beads.
Put each beads in a well on top of a sensor.
Detect the bases of the fragments by a method such as pyrosequencing.
Put the puzzle back together by using an assembly algorithm which detects overlaps in primer DNA.
Then describe the slides below for each of the points mentioned above.

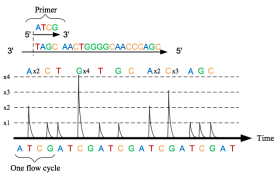


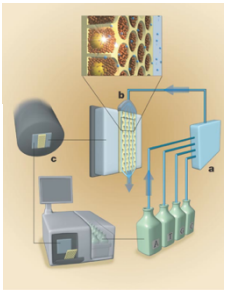


Solution

DNA sequencing technology

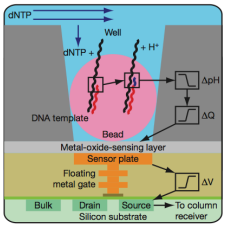
- Pyrosequencing
- Insert bases sequentially and see if there is a match.

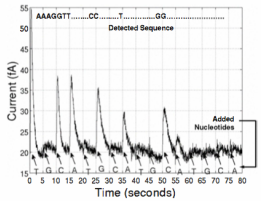




Semiconductor Sequencing

- Use an ISFET array to detect a change in pH when there is a match.





[Total 10 points]

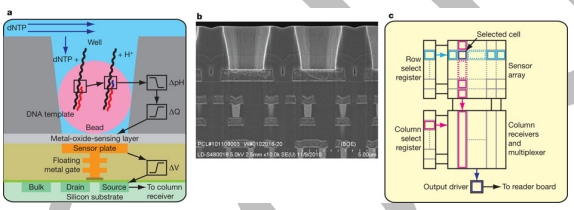
b) DNA sequencing has been revolutionised with the introduction of semiconductor sequencing. This uses a CMOS chip with millions of chemical sensors to conduct an in-parallel investigation of the DNA. You are required to design this CMOS chip describing the following:

- The choice of your sensor to detect DNA and the mechanism of sensing.
- The sensors operating characteristics, showing equations describing its function.
- The choice of instrumentation for this sensor.
- The circuit diagram implementation for an array to scale the design to millions of sensors.

Student needs to identify that this is an ISFET and explain its operation, show the method of instrumentation with justification and then sketch the circuit diagram of a 2x2 array taught in the lecture.

ISFETs for Genetic Sequencing

- Sensor and chip architecture



Equivalent Macromodel

Linear Region: ($V_{DS}<V_{GS}-V_t$)

Drain Current: $I_{DS(si)} = 2\beta(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2}$

Saturation Region: ($V_{DS}>V_{GS}-V_t$)

Drain Current: $I_{DS(si)} = \beta(V_{GS} - V_T)^2(1 + \lambda V_{DS})$

Transconductance: $g_{m(si)} = 2\sqrt{\beta \cdot I_D}$

Current Factor: $\beta = \frac{KW}{2L}$

ISFET Threshold voltage: $V_T = V_{TH(MOS)} + V_{chem}$

Chemical Relation: $V_{chem} = \gamma + 2.3\alpha U_t pH$

γ is a grouping of chemical potentials, α is a sensitivity factor (0-1), U_t is the thermal voltage (26mV).

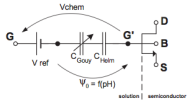
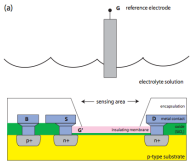


Figure 4.6: ISFET Behavioural Macromodel



Solutions

pH sensitive ISFETs

- Threshold voltage shifts with pH concenration.
- 1 pH unit causes a change of 59mV.
- This is known as the Nernst Potential - 59mV/pH

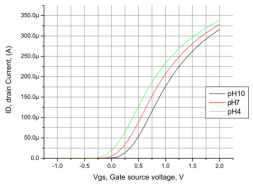
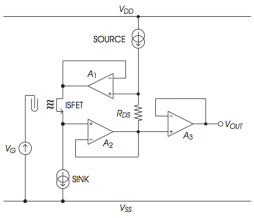


Figure 4.3: Measured I_D-V_{GS} characteristic for variable pH.

ISFET readouts – Opamp based

- Constant charge method (CCM) is the most popular.
- Fix V_G, I_D, V_{DS} to operate at a fixed working point.
- V_S then tracks changes in V_T.
- Opamp A_2 tracks changes in VS and through feedback of A_1, sets V_D to ensure V_{DS} is always constant.



Circuit diagram of an ISFET source-and-drain follower circuit

Large Scale Integration

- ISFETs are inherently CMOS compatible and can be used to make large scale sensing arrays.
- Scaling is dictated by Moore's law just like the MOSFET.

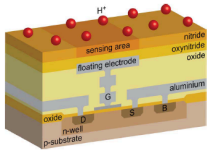


Fig. 1. 3-D cross section through a pH-sensitive p-type ISFET with a floating-gate electrode.

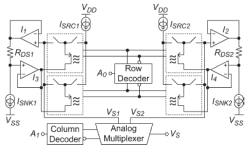


Fig. 3. Circuit diagram of a 2 × 2 array including ISFETs, source-and-drain followers, current sources and sinks, and readout circuitry.

[Total 10 points]

Solutions